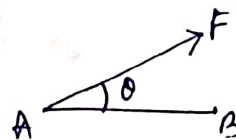


Work, Power & Energy

§ Work:- If the force is constant the work done by it is defined as the product of the magnitude of the force and the distance of the force through which the point of application of the force moves in the direction of the force.

Work = Force $\times$ Displacement of the point of application along the line of action of the force.

Let the force $\vec{F}$ move its point of application ~~along the line of action of~~ from A to B where $AB = s$



$$\therefore \text{Work} = s \cdot F \cos \theta$$

§ Note 1:- When $\theta = 0$, i.e. the displacement is along the direction of the force, Work = force $\times$ displacement

§ Note 2:- When $\theta = \pi$, i.e. the displacement is in the opposite direction of the force, work done is negative & work is said to be done against the force.

Unit of Work:-

1. F.P.S Method Foot-Poundal
2. M.K.S " Kilogram-Meter
3. C.G.S " Erg.

Power:- Power is defined as the rate of work.

$$\text{Power} = \frac{W}{t} = FV = \text{Force} \times \text{Velocity}$$

Unit of Power:-

1. In F.P.S Foot-Poundal per second
2. In C.G.S erg per second

$$1 \text{ H.P.} = 1 \text{ Joule per second} = 10^7 \text{ erg per second} \\ = 550 \text{ foot-Pounds per second}$$

& Note 1:-

If an engine moving with a Velocity v ft/sec exerts a force F lbs-wt in the direction of motion, then the work done by the engine is Fv ft.-lbs.

If H be the Horse power of the engine then $550H = Fv$

$$\therefore \boxed{F = \frac{550H}{v} \text{ lbs.-wt}}$$

& Note 2:- The Dimension of Power

(= mass $\times$ Acceleration $\times$ Velocity)

$$= ML^2T^{-3}$$

& Kinetic Energy:- Kinetic energy of a body is the energy it possesses by virtue of its motion and is measured by the amount of work which the body can do against any resisting force before its velocity is destroyed.

& Potential Energy:- Potential energy of a body is its capacity for doing work on account of its position or configuration relative to other bodies and is measured by the amount of work which the body can do in passing from its present position or configuration to some standard position or configuration.

& The K.E. of a particle of mass m moving with a velocity v is $\frac{1}{2}mv^2$.

& Principle of Energy:-

The change in K.E. of a body is equal to the work done by the acting force.

1. 2020:- In C.G.S. system the absolute unit of work is
i) Foot-Poundal ii) Erg iii) ~~the~~ Horse Power iv) None
An. Erg.

2. 2020:- The work done in raising a mass of 2 kg to a height of 5 meter is

i) 98 joules ii) 25 joules iii) 97 joules iv) 96 joules

An. Work done = mgh

$$= 2 \times 9.8 \times 5 \text{ joules}$$

$$= 98 \text{ joules}$$

3. 2020:- The work done by gravity on a stone of mass 80 gms during the 8th second of its fall from rest is

$$[g = 980 \text{ cm/s}^2]$$

i) 588000 erg ii) 5880 ergs iii) 588 ergs iv) None.

An. Distance fallen in the 8th second

$$= \text{Distance fallen in 8 sec} - \text{Distance fallen in 7 sec.}$$

$$= \frac{1}{2} \cdot g \cdot 8^2 - \frac{1}{2} \cdot g \cdot 7^2$$

$$= \frac{1}{2} \cdot g \cdot (64 - 49) = \frac{1}{2} \cdot g \cdot 15 \text{ cm}$$

$$= \frac{1}{2} \times 980 \times 15 \text{ cm.}$$

$\therefore$ Work done by gravity in the 8th second $g = 980 \text{ cm/s}^2$

$$= \text{force} \times \text{distance}$$

$$= m \times g \times \frac{1}{2} \times 980 \times 15 \text{ erg}$$

$$= 80 \times 980 \times \frac{1}{2} \times 980 \times 15$$

$$= 576240000 \text{ ergs.}$$

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$$= \text{force} \times \text{distance}$$

$$= m \times g \times \frac{1}{2} \times 980 \times 15 \text{ erg}$$

$$= 80 \times 980 \times \frac{1}{2} \times 980 \times 15$$

$$= 576240000 \text{ ergs.}$$

4. 2020:- The potential energy of a particle of mass m at a height h above the Earth's surface is $\frac{1}{2} mgh$ $\frac{1}{2} mg$ mg gh $None$.
 Ans. mgh .

5. 2020:- Horse Power =

i) 746.3 watts $\frac{1}{2}$ 750 watts $\frac{1}{2}$ 740 watts $None$

Ans. $1 \text{ foot} = 30.48 \text{ cm}$ $\times g = 32.2 \text{ ft/sec}^2$
 $1 \text{ Pound} = 453.6 \text{ gm}$.

$$1 \text{ ft-Poundal/sec} = 453.6 \times (30.48)^2 \text{ erg/sec}$$

$$1 \text{ H.P.} = 550 \text{ ft-lb/sec}$$

$$= 550 \times 32.2 \text{ foot-Poundal/sec}$$

$$= 550 \times 32.2 \times 453.6 \times (30.48)^2 \text{ erg/sec}$$

$$= 746.3 \times 10^7 \text{ erg/sec}$$

$$= 746.3 \text{ Joules/sec}$$

$$= 746.3 \text{ Watt.}$$

6. 2020:- The dimension of Power is

i) ML^2T^{-2} $\frac{1}{2}$ ML^2T^{-1} $\frac{1}{2}$ ML^2T^{-3} $None$

7. For uniformly accelerated motion of a particle of ~~constant~~ constant mass its velocity at two instants are 3 ft/sec & 5 ft/sec. Find the work done by the forces during this interval of time. (82)

Ans. By the principle of Energy

$$\text{Work done by the forces} = \text{Change in K.E.}$$

$$= \left(\frac{1}{2} \cdot 1.5^2 - \frac{1}{2} \cdot 1.3^2 \right) \text{ ft-lb}$$

$$= 8 \text{ ft-lb.}$$

8. At what height would the K.E. of a falling particle be equal to half its potential energy?

Ans. Let h be the total height & x be the velocity of the falling particle at height x , when the K.E. is equal to half the P.E.

Since the body falls through a distance $(h-x)$

$$v^2 = 2g(h-x)$$

If m be the mass of the particle, at a height x ,

$$\text{its K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m \times 2g(h-x) = mg(h-x)$$

$$\text{and P.E.} = mgx$$

By the condition of the problem,

$$mg(h-x) = \frac{1}{2}mgx$$

$$\text{or, } h = \frac{3x}{2}$$

$$\text{or, } x = \frac{2h}{3}$$

9. What is the horse power of the engine which keeps a train moving with velocity v ft/sec against a resistance of R pounds?

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Ans. Let H be the horse power of the engine and P be the pull exerted by the engine.

As the velocity is uniform, acceleration $f=0$

$$\therefore P - R = \text{net resultant force} = mf = 0 \quad \text{Since } f=0$$

$$\therefore P = R$$

$$\text{Power} = P \times v = R \cdot v \text{ ft-lbs/sec}$$

$$\text{Required H.P.} = \frac{Rv}{550}$$

10. A motor boat of 40 HP working at full speed moves at 20 m.p.h. Find the resistance of water. 86

$$\text{Ans. Velocity of the boat} = \frac{20 \times 1760 \times 3}{60 \times 60} \text{ ft/sec} = \frac{88}{3} \text{ ft/sec}$$

If R lbs-wt be the resistance of water.

$$\text{Work done in a second} = 40 \times 550 = \frac{88}{3} \times R \quad \therefore R = \frac{3 \times 40 \times 550}{88} = 750 \text{ lbs-wt}$$

